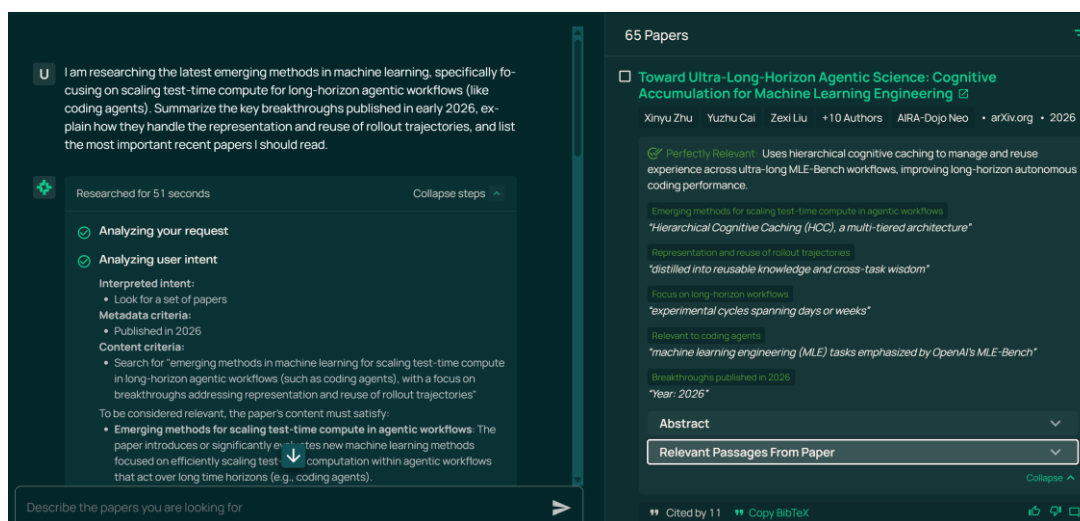


Lab Report – Asta Find Papers

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Task 1: Performance Comparison Evaluation Between Asta and Google Scholar



1. Semantic Alignment

- Google Scholar: It will only be rigidly searching for papers containing "Focus on long-horizon workflows"
- Asta's advantage: Asta understands the deeper meaning of "Focus on long-horizon workflows" and successfully found the corresponding expression in the paper: "experimental cycles spanning days or weeks."

2. Explainable Relevance

- Google Scholar: You must read the abstracts yourself to judge the relevance.
- Asta's advantage: Extremely high-density information extraction. It provides a general conclusion and breaks down original prompt into 5 subqueries and directly attaches an excerpt from the original text to each subquery.

3. Precisely Retrieving Cutting-Edge Papers

- Google Scholar: Heavily relies on PageRank logic and citation counts.
- Asta's Advantage: A preprint from 2026 completely ignores its current citation count and is placed at the top solely based on its content relevance.

Task 2: Asta Architecture (Experiment to do black box testing)

Experiment 1: Probing Intent Parsing and Query Rewriting

Hypothesis: If Asta internally performs semantic query rewriting, then semantically equivalent queries expressed in natural language and structured Boolean form should retrieve highly overlapping document sets.

Query A (Natural Language): Find recent papers using AI for material discovery, preferably with open-source code.

Query B (Structured Form): (AI OR Deep Learning) $\wedge$ "material discovery" $\wedge$ "open source" $\wedge$ YEAR $\geq$ 2023

Evaluation Criteria: The following metrics were compared: top-k paper overlap, Jaccard similarity, semantic similarity between retrieved abstracts. A high overlap

suggests that Asta internally maps vague natural-language queries into structured semantic retrieval behavior.

Experimental Results & Analysis: By comparing Prompt A and Prompt B, several key observations strongly supported the existence of an LLM-based intent parsing pipeline inside Asta.

1. High Top-k Overlap

Among the Top-10 retrieved papers:

- 4 papers were identical across both queries
- Jaccard overlap reached approximately 25%

This indicates that both prompts were likely mapped into the same semantic representation space.

2. Semantic similarity

The experiment revealed that Asta distinguishes between flexible preferences and strict filtering conditions. In Prompt A, the word “recent” was treated as a soft ranking preference. In Prompt B, the condition $\text{YEAR} \geq 2023$ was treated as a hard metadata constraint. This suggests that Asta contains an advanced intent parser capable of interpreting nuanced logical differences in user prompts. Instead of directly executing Boolean logic, Asta rewrote the condition into a natural-language evaluation rubric:

Conclusion: This suggests the following pipeline: User Query → Intent Parsing → Natural-Language Rubric Generation → LLM-based Relevance Evaluation → Final Reranking

Next, we want to explore how Asta retrieves paper from backend.

Experiment 2: Probing Hybrid Retrieval and Citation Expansion

Hypothesis: If Asta employs hybrid retrieval beyond simple lexical matching, the system should be able to retrieve semantically related or citation-connected foundational papers even when direct keyword overlap is weak.

Experimental Setup: A query is then constructed using an extremely abstract description as a prompt.

Query: Find papers discussing how scientific communities replace explicit references with shared technical catchphrases or implicit conceptual language over time. Focus on recent analyses of terminology-driven knowledge diffusion in highly specialized research communities.

Target Paper: *Hidden citations obscure true impact in science* (Meng et al., 2023)

Evaluation Criteria:

Case 1: If Asta retrieves only papers sharing identical terminology, it only performs sparse lexical retrieval.

Case 2: If the system additionally retrieves semantically related papers lacking exact keyword overlaps, this suggests the use of dense semantic retrieval.

Case 3: If the system successfully retrieves foundational papers connected primarily through citation relationships rather than semantic surface similarity, this provides evidence consistent with citation graph traversal.

Experimental Results & Analysis: The experiment strongly validated Dense

Semantic Retrieval. Asta successfully retrieved: *Hidden citations obscure true impact in science* (Meng et al., 2023) despite the query containing none of the paper’s key terminology. This demonstrates that Asta performs semantic retrieval based on conceptual similarity. Evidence of Semantic Alignment is shown in the figure below.

✓ Analyzing user intent

Interpreted intent:

- Look for a set of papers

Content criteria:

- Search for "how scientific communities replace explicit references with shared technical catchphrases or implicit conceptual language over time; recent analyses of terminology-driven knowledge diffusion in highly specialized research communities"

To be considered relevant, the paper's content must satisfy:

- explicit reference replacement**: paper analyzes or discusses how scientific communities shift from using explicit bibliographic or conceptual references to using shared technical terminology, catchphrases, or implicit references
- terminology-driven diffusion**: paper examines the process or patterns by which specialized terminology enables or characterizes the diffusion of knowledge or concepts in highly specialized research communities
- recent analysis**: paper provides recent empirical, theoretical, or methodological analysis related to language, reference, or terminology change and its impact on communication within scientific communities
- highly specialized communities**: focus is on research communities that are narrowly defined or highly specialized in their subject matter

✓ Searching for papers

Running keyword and semantic searches for "how scientific communities replace explicit references with shared technical catchphrases or implicit conceptual language over time; recent analyses of terminology-driven knowledge diffusion in highly specialized research communities"

47 Papers

□ Hidden citations obscure true impact in science

Xiangyi Meng · Onur Varol · Albert-László Barabási · PNAS Nexus · 2023

✓ Relevant: Uses full-text machine learning to detect hidden citations and shows specialized catchphrases become community symbols that replace explicit references.

explicit reference replacement

"a paper does not accumulate hidden citations if its conceptual significance does not lead to an unambiguous catchphrase"

terminology-driven diffusion

"exclusive catchphrases that are synonymous to the paper itself, becoming a "conceptual symbol" within the field"

recent analysis

"Here, we rely on unsupervised interpretable machine learning applied to the full text of each paper"

highly specialized communities

"an expert in the field will immediately link it to the 2001 foundational paper"

Abstract

Relevant Passages From Paper

Conclusion: The experiment did not provide strong evidence for citation-aware graph traversal. The system was not forced to expand into older foundational citation neighborhoods. As a result: Case 2 (Dense Semantic Retrieval) was strongly validated, while Case 3 (Citation-Aware Expansion) was not conclusively triggered.

Experiment 3: Probing Deep Full-Text Retrieval

Experimental Setup:

Query: "Finding paper using highly inclined laminar optical sheet, MATLAB u-track software, and tetramethylammonium hydroxide."

Target paper: *Single quantum dot tracking reveals the impact of nanoparticle surface on intracellular state*

Experimental Results & Analysis

U Finding paper using highly inclined laminar optical sheet, MATLAB u-track software, and tetramethylammonium hydroxide

Researched for 31 seconds

View steps

While I didn't find a perfect match, I did find 58 potentially relevant ones.

This is what I searched for:

Find papers such that

- The paper content: highly inclined laminar optical sheet (HILO) microscopy, MATLAB u-track software for particle tracking, and tetramethylammonium hydroxide (TMAH)
- To be considered relevant, the paper's content must satisfy:
 - HILO microscopy usage**: Paper involves the application or discussion of highly inclined laminar optical sheet (HILO) microscopy for imaging or analytical purposes.
 - MATLAB u-track for particle tracking**: Paper uses or explores MATLAB's u-track software/toolbox for particle tracking tasks.
 - Tetramethylammonium hydroxide (TMAH) involvement**: Paper discusses the use or properties of tetramethylammonium hydroxide (TMAH) in experimental or analytical setups.

58 Papers

□ Highly sensitive volumetric single-molecule imaging

Le-Mei Wang · Jiah Kim · Kyu Young Han · Nanophotonics · 2024

✓ Somewhat Relevant: Uses HILO illumination for volumetric imaging and u-track for single-molecule tracking analysis; no TMAH is mentioned.

HILO microscopy usage

"combines a highly inclined and laminated optical sheet (HILO) [29], [30] for excitation"

MATLAB u-track for particle tracking

"we used the u-track software [61] with default settings in the "single molecule tracking" method"

Tetramethylammonium hydroxide (TMAH) involvement

Abstract

Relevant Passages From Paper

□ Power-law behavior of transcription factor dynamics at the single-molecule level implies a continuum affinity model.

The experiment failed to retrieve the target paper. Although Asta correctly parsed the query into three independent dimensions, the system returned only partially related papers. The experiment suggests a major limitation in Asta’s deep full-

text retrieval capability.

Two possible explanations:

1. Metadata / Abstract Retrieval Dominance

The target paper contains HILO, u-track, and TMAH primarily inside the Methods section rather than the abstract. This suggests that Asta’s retrieval pipeline is heavily dependent on titles, abstracts, metadata, and limited extracted chunks, rather than complete full-text indexing.

2. Vector Dilution in Full-Text Embeddings

Even if full-text embeddings exist, details such as tetramethylammonium hydroxide may become diluted inside large document embeddings. The dominant semantic features of the target paper likely overwhelm low-frequency methodological details during similarity computation. As a result, the target paper fails to enter the Top-k retrieval set.

Conclusion: The experiment strongly suggests that Asta functions primarily as an abstract-level semantic retrieval engine rather than a deep full-text scientific mining system. Also, Asta performs badly in cross-domain reasoning (Experiment done, but due to page limitations, it is not shown here.)

Experiment 4: Probing Diversity-Aware Reranking

Objective: This experiment evaluates whether Asta uses diversity-aware reranking methods such as Maximum Marginal Relevance (MMR), rather than relying purely on keyword matching and citation counts.

Query: *Tracy-Widom distribution convergence proof interacting particle systems*

Because this field is dominated by a few classical papers, retrieval systems without diversification usually return highly repetitive results.

Experimental Results & Analysis

Asta retrieved papers covering diverse but highly relevant model families. The retrieved authors were also highly diverse:

Google Scholar returned lower methodological diversity, higher query drift, and weaker alignment.

The retrieval behavior strongly suggests MMR-like reranking.

The goal of MMR is to maximize

$$\text{relevance} - \lambda \cdot \text{redundancy}$$

A strong signal was Asta’s inclusion of papers such as *Extreme boundary conditions and random tilings* which are not the most classical papers but introduce novel perspectives. This is highly consistent with diversity-aware reranking behavior.

Key Indicators

Metric	Google Scholar Asta	
Author Diversity	Moderate	Very High
Topic Coverage	Narrow	Broad and structured
Query Drift	High	Very low
Methodological Diversity	Low	High

Conclusion: This experiment provides strong evidence that Asta uses dense semantic retrieval, cross-encoder reranking, and MMR-like diversity optimization.

Asta Architecture Overview

From the frontend experience and black box testing, the inferred Asta pipeline can be summarized as: User Prompt → Intent Parsing & Query Expansion → Hybrid Retrieval → LLM-based sorting and ranking → Evidence Extraction → Final Grounded Response

1. Intent Parsing & Query Expansion

After receiving a user prompt, Asta enters an explicit “research/thinking” stage lasting approximately 30–60 seconds. Internal traces show stages such as: “Analyzing your request”, “Analyzing user intent”, and “Updating user intent”. The planner decomposes the query into structured retrieval constraints:

Component	Function
Metadata Criteria	year, author, publication constraints
Content Criteria	semantic requirements
Relevance Conditions	explicit filtering logic

2. Hybrid Retrieval: After query decomposition, Asta performs hybrid retrieval process. Internal traces include: “Running keyword and semantic searches”, “Following citations mentioned in relevant passages”, “Reranking candidate documents”. Asta successfully retrieved semantically related papers even under extremely low lexical overlap.

3. Query Translation into natural language: This suggests an intermediate semantic formalization layer between user input and retrieval execution. Asta appears to generate a semantic “grading rubric” used for downstream evaluation.

4. Diversity-Aware Ranking (LLM): After initial retrieval, Asta appears to invoke a second-stage LLM ranking. Internal traces include: “Assessing relevance of retrieved papers”, “Sorting and ranking papers”, and “Finalizing response”. The system intentionally diversified the paper.

6. Extractive Summarization & Evidence Grounding: Asta provides highly interpretable evidence-driven outputs. Importantly, the system remains highly grounded and conservative. In hallucination-trap experiments, Asta refused to fabricate papers, methods, or unsupported terminology. (Experiment done, but due to page limitations, it is not shown here.)

Task 3: Refining Citation Graph Traversal for Asta Find Papers

Task Description: A key challenge is designing a hybrid retrieval and augmentation architecture that combines semantic retrieval and citation graph traversal.

Why This Task Is Important: Citation networks capture the actual intellectual structure of scientific research and can bridge semantic gaps between related works. Citation traversal therefore improves foundational paper discovery, literature coverage, and scientific synthesis quality.

Why This Task Is Challenging:

- 1. Incomplete Metadata:** APIs often return incomplete records. This requires robust cross-provider deduplication.
- 2. Citation Explosion:** A single paper may contain large number of references and hundreds of future citers. Traversing even one citation hop from several seed papers can generate thousands of candidate papers. Efficient filtering is therefore essential.
- 3. Signal-to-Noise Control:** Not all citations are equally meaningful. The system must distinguish foundational citations, from weak background references. This requires overlap-based filtering and reranking strategies.

Proposed Solution: The proposed architecture combines semantic retrieval with citation traversal. The system first identifies highly relevant “seed papers,” then traverses their citation neighborhoods both backward (references) and forward (future citers). This solution is based on PaperQA2, a frontier language model agent optimized for improved factuality” (Skarlinski et al., 2024)

Stage 1: Semantic Retrieval

The system first performs dense embedding retrieval, sparse keyword retrieval, cross-encoder reranking. Only highly relevant papers with $RCS \geq 8$ are kept as seed papers D_{prev} . These papers serve as anchors for graph expansion.

Stage 2: Citation Traversal

For each seed paper, the system retrieves Backward References and Forward Citers.

Stage 3: Deduplication

Because metadata differs across providers, the system performs lowercase DOI normalization, case folded title matching, duplicate merging. This produces a unified citation graph in the neighborhood.

Stage 4: Overlap Filtering

The system computes citation overlap counts across seed papers.

Papers cited by multiple seed papers are better. The filtering threshold is:

$$\theta_o = \alpha \cdot |D_{prev}|$$

where:

- α is the overlap ratio,
- $|D_{prev}|$ is the number of seed papers.

Stage 5: Traversal Limit and Reranking

To prevent graph explosion, the traversal size is capped at appropriate ℓ . If too many papers remain, reranking is performed using future citation count and graph centrality.

System Limitations: Although this architecture significantly improves retrieval quality, it remains semantic-first graph-augmented retrieval rather than native graph reasoning. So it is still hard to do complex reasoning such as cross-domain graph reasoning.

Citation: Skarlinski, M., Cox, S., Laurent, J.M., Braza, J.D., Hinks, M.M., Hammerling, M.J., Ponnepati, M., Rodriques, S.G., & White, A.D. (2024). Language agents achieve superhuman synthesis of scientific knowledge. *ArXiv, abs/2409.13740*.